

Queueing Theory

Md Irtiaz Kabir

What Is a Queueing Model?

A **queueing model** describes a facility where customers arrive randomly, wait if necessary, get served, and then leave.

We want to answer questions like:

- On average, how many customers are in the system (or in the queue)?
- On average, how long does a customer spend in the system (or waiting)?

Why this matters

These questions show up everywhere: bank tellers, call centers, web servers, hospital wards, manufacturing lines, network routers. . . any place where “jobs” compete for a scarce resource.

Four fundamental quantities

- L — the average number of customers in the **system**.
- L_Q — the average number of customers waiting in **queue**.
- W — the average time a customer spends in the **system**.
- W_Q — the average time a customer spends waiting in **queue**.

Almost everything we do in this chapter is aimed at computing these four numbers (or their state-by-state refinements) for various models.

The Fundamental Cost Identity

Imagine that entering customers are forced to pay money according to some rule.
Then, quite generally,

Basic cost identity

average rate at which the system earns = $\lambda_a \times$ average amount an entering customer pays,

where λ_a is the average rate at which customers **enter** the system,

$$\lambda_a = \lim_{t \rightarrow \infty} \frac{N(t)}{t}, \quad N(t) = \text{number of arrivals by time } t.$$

This identity holds for **almost every** queueing model, regardless of the arrival process, the number of servers, or the queue discipline.

Special Cases: Little's Law and Its Relatives

By choosing the cost rule cleverly, the same identity yields several famous formulas at once.

Pay \$1 per unit time while in the system

$$L = \lambda_a W \quad (\text{Little's Law})$$

Pay \$1 per unit time while in queue

$$L_Q = \lambda_a W_Q$$

Pay \$1 per unit time while in service

$$\text{average number in service} = \lambda_a \mathbb{E}[S]$$

where $\mathbb{E}[S]$ is the average time a customer spends being served.

Steady-State Probabilities

Let $X(t)$ denote the number of customers in the system at time t , and define

$$P_n = \lim_{t \rightarrow \infty} \mathbb{P}\{X(t) = n\}, \quad n \geq 0.$$

Interpretation

P_n is the long-run **proportion of time** the system contains exactly n customers.

Two other natural sets of probabilities describe what **arrivals** and **departures** experience:

a_n = proportion of arrivals that find n in the system,

d_n = proportion of departures that leave behind n in the system.

Do Arrivals, Departures, and Time Averages Agree?

In general, P_n , a_n , and d_n need **not** be equal — a system can look different depending on *who* is looking at it (a random outside observer, an arriving customer, or a departing customer).

Two questions we will answer

1. Is it always true that $a_n = d_n$?
2. When is it true that $P_n = a_n$?

Proposition — Arrivals and Departures See the Same Rates

Proposition

In any system where customers arrive and depart one at a time,

the rate at which arrivals find n = the rate at which departures leave n ,

and consequently $a_n = d_n$ for every n .

This holds regardless of the arrival process, the service mechanism, or the number of servers — it is a purely **bookkeeping** fact.

Intuition — Why $a_n = d_n$

An arrival **finds** n in the system exactly when the system count jumps from n to $n + 1$.
A departure **leaves behind** n exactly when the count drops from $n + 1$ to n .

- Between any two up-crossings $n \rightarrow n+1$, there must be a down-crossing $n+1 \rightarrow n$ (and conversely) — the count cannot cross the “ $n \leftrightarrow n+1$ ” boundary twice in a row in the same direction.
- So over any long interval, the number of up-crossings and down-crossings differ by at most 1.

Hence the two rates are equal, and dividing each by its own overall rate (arrival rate vs. departure rate — which coincide in a stable system) gives $a_n = d_n$.

Proposition (Poisson Arrivals See Time Averages)

If customers arrive according to a **Poisson process**, then

$$P_n = a_n \quad \text{for every } n.$$

This is the celebrated **PASTA** principle: a Poisson arrival sees the system exactly as a random outside observer would.

Intuition — Why PASTA Holds

- A Poisson process has **independent increments**: knowing that an arrival occurs at time t tells us nothing about what happened *before* t . So the state an arrival sees at t has the same distribution as the unconditional state at t — namely, P_n .
- **Second argument**: Poisson arrivals occur at rate λ *no matter what state the system is in*. So the rate at which arrivals find state n is λP_n , and dividing by the overall arrival rate λ gives back P_n .

Contrast

If interarrival times were *not* exponential, knowing an arrival just occurred can reveal information about the recent past (e.g., “no arrivals in the last unit of time”), and PASTA can fail.

Recall — What Does λ Mean?

Before we build the $M/M/1$ queue, let's recall what λ stands for.

λ is a rate

If $T \sim \text{Exponential}(\lambda)$, then

$$\mathbb{E}[T] = \frac{1}{\lambda}, \quad f(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

λ is the number of events **per unit time**, on average — a bigger λ means events happen *faster* (shorter average gaps between them).

In a **Poisson process** with rate λ , the times between successive events (interarrival times) are IID $\text{Exponential}(\lambda)$, and the number of events in an interval of length t is $\text{Poisson}(\lambda t)$.

The $M/M/1$ Queueing System — Setup

- Customers arrive according to a **Poisson process** with rate λ — interarrival times are IID Exponential(λ).
- There is a **single server**. An arriving customer enters service immediately if the server is free, and otherwise joins the queue.
- Service times are IID Exponential(μ).

Why “ $M/M/1$ ”?

- first M = **Memoryless arrival** process,
- second M = **Memoryless service** process,
- $1 = 1$ server.

Intuition — Where Do the Balance Equations Come From?

Write $P_n(t) = \mathbb{P}\{X(t) = n\}$. Quite generally,

rate of change of $P_n(t)$ = rate of *entering* n – rate of *exiting* n .

A concrete picture

Suppose the system starts empty: $P_0 = 1$ and $P_n = 0$ for $n \geq 1$. As soon as one customer arrives, the system starts spending time in state 1 too — say a moment later the time is split $(0.5, 0.5)$ between states 0 and 1. So P_0 has **decreased** and P_1 has **increased**: the distribution is still *moving*.

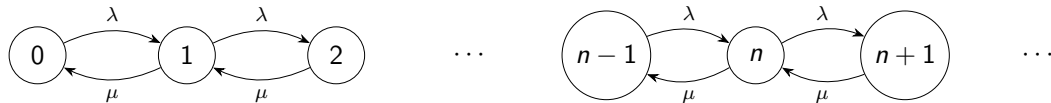
The long-run, stable condition

If the system settles into a long-run stable condition (its limiting, Markov-chain proportions), the P_n 's stop changing, so the rate of change is 0:

rate of entering n = rate of exiting n — the **balance equation**.

If this fails to hold, the system simply has **not yet** reached its long-run proportions.

Balance Equations for $M/M/1$



From state 0, the system can only leave via an arrival, and can only be entered via a departure from state 1. From state $n \geq 1$, it can leave via an arrival *or* a departure (combined rate $\lambda + \mu$).

Balance equations

$$\lambda P_0 = \mu P_1, \quad (\lambda + \mu)P_n = \lambda P_{n-1} + \mu P_{n+1}, \quad n \geq 1.$$

Solving the Balance Equations

Starting from the balance relation (for $n \geq 1$) and dividing by μ :

$$\mu(P_{n+1} - P_n) = \lambda(P_n - P_{n-1}) \implies P_{n+1} - P_n = \frac{\lambda}{\mu}(P_n - P_{n-1})$$

Let $D_n := P_{n+1} - P_n$. The recurrence becomes

$$D_n = \frac{\lambda}{\mu} D_{n-1} \implies D_n = \left(\frac{\lambda}{\mu}\right)^n D_0,$$

From the state-0 equation $\lambda P_0 = \mu P_1$:

$$D_0 = P_1 - P_0 = \frac{\lambda}{\mu} P_0 - P_0 = \left(\frac{\lambda}{\mu} - 1\right) P_0.$$

Substituting back,

$$P_n - P_{n-1} = \left(\frac{\lambda}{\mu}\right)^{n-1} \left(\frac{\lambda}{\mu} - 1\right) P_0.$$

Solving the Balance Equations — Telescoping

Summing telescopically to find P_n :

$$\begin{aligned}P_n &= P_0 + \sum_{k=0}^{n-1} (P_{k+1} - P_k) \\&= P_0 + \left(\frac{\lambda}{\mu} - 1\right) P_0 \sum_{k=0}^{n-1} \left(\frac{\lambda}{\mu}\right)^k \\&= P_0 + \left(\frac{\lambda}{\mu} - 1\right) P_0 \times \frac{(\lambda/\mu)^n - 1}{\lambda/\mu - 1} \\&= \left(\frac{\lambda}{\mu}\right)^n P_0.\end{aligned}$$

Solving the Balance Equations — Normalization

Normalizing with $\sum_{n=0}^{\infty} P_n = 1$:

$$P_0 \sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n = 1 \quad \Rightarrow \quad P_0 \cdot \frac{1}{1 - \lambda/\mu} = 1 \quad \Rightarrow \quad P_0 = 1 - \frac{\lambda}{\mu}.$$

Result

$$\therefore P_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), \quad n \geq 0.$$

The Stability Condition $\rho = \lambda/\mu < 1$

The geometric series $\sum_n (\lambda/\mu)^n$ only converges when $\lambda/\mu < 1$. We call

$$\rho = \frac{\lambda}{\mu}$$

the **traffic intensity** — and we can **prove** it equals the **server utilization**, the long-run proportion of time the server is busy (not idle):

$$\text{server utilization} = 1 - P_0 = 1 - \left(1 - \frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu} = \rho.$$

If $\lambda \geq \mu$?

No limiting probabilities exist: the queue grows without bound. The server simply cannot keep up with arrivals in the long run.

Intuition — Why We Need $\lambda < \mu$

Suppose $\lambda > \mu$. By time t , the expected number of **arrivals** is λt .

Even if the server were *never* idle, the number it could possibly **serve** by time t is a Poisson process of rate μ , so at most about μt customers are served.

Consequence

The number of customers in the system is at least $\lambda t - \mu t = (\lambda - \mu)t \rightarrow \infty$. The queue explodes — exactly the intuitive condition that **the mean service time must be less than the mean interarrival time**.

Performance Measures: L , W , L_Q , W_Q

Using $L = \sum_n nP_n$ together with the geometric-series identity $\sum_{n \geq 1} np^n = p/(1-p)^2$:

$$L = \frac{\lambda}{\mu - \lambda}$$

Then Little's Law ($W = L/\lambda$) and the cost identities give the rest:

$$W = \frac{1}{\mu - \lambda}, \quad W_Q = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}, \quad L_Q = \lambda W_Q = \frac{\lambda^2}{\mu(\mu - \lambda)}.$$

Now suppose the system can hold **at most** N customers: an arrival that finds N customers already present simply does not enter.

- Arrivals: Poisson, rate λ .
- Service: single server, rate μ .
- State space: $\{0, 1, \dots, N\}$ — bounded.

Because the queue size can never exceed N , we will see that a stability condition like $\lambda < \mu$ is **no longer needed**.

Balance equations

$$\begin{aligned}\lambda P_0 &= \mu P_1, \\ (\lambda + \mu)P_n &= \lambda P_{n-1} + \mu P_{n+1}, \quad 1 \leq n \leq N-1, \\ \mu P_N &= \lambda P_{N-1}.\end{aligned}$$

The equation at state 0 and the middle states are exactly as before. State N is different: it can only be **left** via a departure (arrivals are turned away), and can only be **entered** via an arrival from state $N-1$.

Solving for P_n

Exactly as before, $P_n = (\lambda/\mu)^n P_0$ for $n = 1, \dots, N$ — but now the normalization is a **finite** sum:

$$1 = P_0 \sum_{n=0}^N \left(\frac{\lambda}{\mu}\right)^n = P_0 \cdot \frac{1 - (\lambda/\mu)^{N+1}}{1 - \lambda/\mu}.$$

Result

$$P_n = \frac{(\lambda/\mu)^n (1 - \lambda/\mu)}{1 - (\lambda/\mu)^{N+1}}, \quad n = 0, 1, \dots, N.$$

Intuition

Because the state space is bounded, the sum is automatically finite — **any** $\lambda, \mu > 0$ works, even $\lambda \geq \mu$.

L and W — a Subtlety About “Customers”

As before, $L = \sum_{n=0}^N nP_n$ (a finite sum). Carrying out the sum gives

$$L = \frac{\lambda[1 + N(\lambda/\mu)^{N+1} - (N+1)(\lambda/\mu)^N]}{(\mu - \lambda)[1 - (\lambda/\mu)^{N+1}]}$$

For W , we must be careful about *which* arrival rate to use:

- If we count **every** potential customer (including those turned away, who spend 0 time in the system): $\lambda_a = \lambda$.
- If we only count customers who **actually enter**: since a fraction P_N of arrivals are blocked, $\lambda_a = \lambda(1 - P_N)$.

Little's Law still applies

$$W = \frac{L}{\lambda_a},$$

but the two choices of λ_a give **different** answers to “what is the average wait?” — make sure the question is precise.

A Shoe Shine Shop — Setup

A shop has **two chairs**, each with its own server. A customer first goes to chair 1; when done there, she moves to chair 2 if it is free, or else waits in chair 1 until it frees up.

- Potential customers arrive Poisson, rate λ — but a customer only **enters** if chair 1 is empty (even if chair 2 is occupied!).
- Chair 1 and chair 2 have independent exponential service rates μ_1 and μ_2 .

We want: (a) the proportion of potential customers who enter; (b) the mean number L in the system; (c) the mean time W an entering customer spends; (d) the proportion of entering customers who become **blockers** (finish at chair 1 but must wait for chair 2).

Choosing the Right State Space

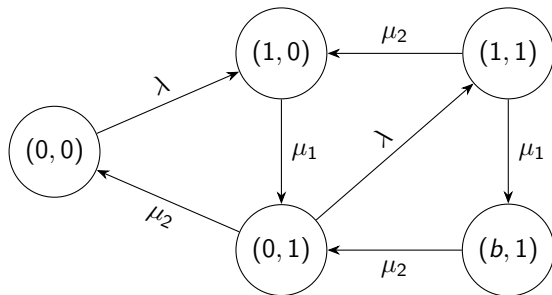
Just counting customers is **not enough** — we also need to know *which chair* each customer occupies, and whether a finished customer is stuck waiting.

Five states

- $(0, 0)$ — system empty.
- $(1, 0)$ — one customer, in chair 1.
- $(0, 1)$ — one customer, in chair 2.
- $(1, 1)$ — two customers, both being served.
- $(b, 1)$ — two customers; chair 1's customer is **done** but blocked, waiting for chair 2.

Note that in state $(b, 1)$, the “blocking” customer in chair 1 prevents any new arrival from entering, even though they are not being actively served.

Transition Diagram



Each arrow shows a possible transition and the rate at which it occurs — exactly the birth–death picture, but on a richer state space.

Balance Equations for the Shoe Shine Shop

Rate leaves = rate enters, for each state

$$\lambda P_{00} = \mu_2 P_{01}$$

$$\mu_1 P_{10} = \lambda P_{00} + \mu_2 P_{11}$$

$$(\lambda + \mu_2) P_{01} = \mu_1 P_{10} + \mu_2 P_{b1}$$

$$(\mu_1 + \mu_2) P_{11} = \lambda P_{01}$$

$$\mu_2 P_{b1} = \mu_1 P_{11}$$

together with $P_{00} + P_{10} + P_{01} + P_{11} + P_{b1} = 1$.

These can be solved for the limiting probabilities — the algebra is routine but the resulting expressions are messy, so we work directly in terms of the P 's.

Answering the Questions — (a) and (b)

(a) Proportion who enter

A potential customer enters only from states $(0, 0)$ or $(0, 1)$, so this proportion is $P_{00} + P_{01}$.

(b) Mean number in system

Since there is 1 customer whenever the state is $(1, 0)$ or $(0, 1)$, and 2 customers whenever it is $(1, 1)$ or $(b, 1)$,

$$L = P_{01} + P_{10} + 2(P_{11} + P_{b1}).$$

Answering the Questions — (c) and (d)

(c) Mean time in system

By Little's Law, with $\lambda_a = \lambda(P_{00} + P_{01})$,

$$W = \frac{L}{\lambda_a}.$$

(d) Proportion who are blockers

A customer becomes a blocker exactly when a service completes at chair 1 while the state is $(1, 1)$. So

$$\pi_b = \frac{\text{rate of chair-1 completions in state } (1, 1)}{\lambda_a} = \frac{\mu_1 P_{11}}{\lambda(P_{00} + P_{01})}.$$

Tandem (Sequential) Queues

Consider two servers in series: customers arrive Poisson rate λ at server 1; after being served there, they join the queue at server 2. Each server has infinite waiting room and exponential service rate μ_i , $i = 1, 2$.

State

(n, m) = number of customers at server 1 and at server 2, respectively.

The balance equations relate $P_{n,m}$ to its neighbors exactly as before, but now in **two dimensions**.

Balance Equations for the Tandem Queue

$$\lambda P_{0,0} = \mu_2 P_{0,1}$$

$$(\lambda + \mu_1) P_{n,0} = \mu_2 P_{n,1} + \lambda P_{n-1,0}, \quad n > 0$$

$$(\lambda + \mu_2) P_{0,m} = \mu_2 P_{0,m+1} + \mu_1 P_{1,m-1}, \quad m > 0$$

$$(\lambda + \mu_1 + \mu_2) P_{n,m} = \mu_2 P_{n,m+1} + \mu_1 P_{n+1,m-1} + \lambda P_{n-1,m}, \quad n, m > 0$$

Rather than grinding through this system directly, we **guess** a solution and verify it.

Guess-and-Verify: A Product-Form Solution

Server 1 behaves exactly like an isolated $M/M/1$ queue with rates λ, μ_1 . What does server 2 face?

Key fact

The departure process of an $M/M/1$ queue is itself a **Poisson process** with rate λ — so server 2 also behaves like an isolated $M/M/1$ queue with rates λ, μ_2 .

This suggests the numbers at the two servers are **independent**:

$$P_{n,m} = \left(\frac{\lambda}{\mu_1} \right)^n \left(1 - \frac{\lambda}{\mu_1} \right) \left(\frac{\lambda}{\mu_2} \right)^m \left(1 - \frac{\lambda}{\mu_2} \right)$$

Substituting back confirms this satisfies every balance equation — so it *is* the (unique) limiting distribution.

Intuition — Why Server 2 Sees Arrival Rate λ

Why does server 2 behave like an $M/M/1$ queue with arrival rate λ , rather than μ_1 (server 1's service rate)?

- Stability at server 1 requires $\lambda < \mu_1$: server 1 is **capable** of serving faster than μ_1 on average, but customers simply don't arrive fast enough to make it work that hard.
- So λ , not μ_1 , is the **bottleneck**: in the long run, customers can only *leave* server 1 at the rate they *arrived* — namely λ .
- Hence the long-run rate at which customers depart server 1 (and so arrive at server 2) is exactly λ — which is why server 2 is itself an $M/M/1$ queue with arrival rate λ .

L and W for the Tandem Queue

Since L is additive over servers,

$$L = \frac{\lambda}{\mu_1 - \lambda} + \frac{\lambda}{\mu_2 - \lambda}.$$

By Little's Law,

$$W = \frac{L}{\lambda} = \frac{1}{\mu_1 - \lambda} + \frac{1}{\mu_2 - \lambda}$$

A customer's total time in the system is simply the sum of the time spent at each server, **as if** the two stages were independent $M/M/1$ queues.

General Open (Jackson) Networks

Now generalize to k servers. Customers arrive from **outside** at server i via an independent Poisson process, rate r_i . After service at i , a customer heads to server j with probability P_{ij} , or leaves the system with probability $1 - \sum_j P_{ij}$.

Traffic equations

Let λ_j be the **total** arrival rate to server j (from outside *and* from other servers):

$$\lambda_j = r_j + \sum_{i=1}^k \lambda_i P_{ij}, \quad j = 1, \dots, k.$$

Solve this linear system for $\lambda_1, \dots, \lambda_k$ first — they play the role that λ played for a single queue.

Product-Form Limiting Probabilities

Provided $\lambda_j/\mu_j < 1$ for every j , the numbers of customers at the different servers are **independent**, each distributed as an $M/M/1$ queue:

$$\mathbb{P}\{n_1 \text{ at } 1, \dots, n_k \text{ at } k\} = \prod_{j=1}^k \left(\frac{\lambda_j}{\mu_j}\right)^{n_j} \left(1 - \frac{\lambda_j}{\mu_j}\right)$$

Each server j “looks like” an isolated $M/M/1$ queue with arrival rate λ_j (the *solution* of the traffic equations) and service rate μ_j .

L and W for General Open Networks

Summing over servers,

$$L = \sum_{j=1}^k \frac{\lambda_j}{\mu_j - \lambda_j}.$$

By Little's Law, with $\lambda = \sum_{j=1}^k r_j$ (the **total outside** arrival rate — *not* $\sum_j \lambda_j$, which double-counts internal transfers),

$$W = \frac{L}{\lambda} = \frac{\sum_{j=1}^k \lambda_j / (\mu_j - \lambda_j)}{\sum_{j=1}^k r_j}$$